

CURITIBA, Brazil

WMO: 838400

Lat: 25.528S

Lon: 49.176W

Elev: 911

StdP: 90.85

Time Zone: -3.00 (SAB)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.1	5.2	-1.1	3.8	9.4	1.6	4.7	10.2	9.9	19.7	8.6	20.3	1.7	120	0.390

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	9.2	31.0	20.1	29.9	20.3	28.8	20.2	23.0	26.9	22.4	26.4	21.8	25.8	4.7	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.0	18.7	24.3	21.2	17.7	23.5	20.9	17.4	23.2	73.3	26.8	70.7	26.2	68.6	26.0	26.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.7	7.6	6.6	DB	-0.5	33.0	1.9	0.7	-1.8	33.5	-2.9	33.9	-4.0	34.3	-5.3	34.7	(4)
(5)			WB	-1.0	24.3	1.9	1.0	-2.4	25.0	-3.6	25.5	-4.7	26.0	-6.1	26.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	18.3	21.9	22.2	21.1	19.2	15.8	14.7	14.4	15.5	16.6	18.2	19.3	21.1	(6)	
	DBStd	4.05	2.28	2.28	2.41	2.82	2.84	3.16	3.39	3.57	3.69	3.12	2.96	2.65	(7)	
	HDD10.0	17	0	0	0	0	1	4	7	4	1	0	0	0	(8)	
	HDD18.3	608	3	2	5	23	87	113	126	98	77	42	24	8	(9)	
	CDD10.0	3050	369	342	343	277	181	145	143	174	198	254	278	345	(10)	
	CDD18.3	599	113	111	90	49	9	3	3	11	24	38	53	94	(11)	
	CDH23.3	3469	550	563	434	237	57	33	61	201	258	258	298	518	(12)	
(13)	CDH26.7	848	138	148	103	40	4	1	2	36	87	76	75	139	(13)	
(14)	Wind	WSAvg	3.1	3.2	3.1	3.0	2.8	2.7	2.8	2.9	3.3	3.5	3.6	3.4	(14)	
(15) Precipitation	PrecAvg	1653	225	188	158	94	100	110	121	79	132	146	134	166	(15)	
	PrecMax	2228	490	330	310	224	277	264	283	252	386	306	226	306	(16)	
	PrecMin	1205	69	71	49	16	10	37	7	14	63	27	46	46	(17)	
	PrecStd	244	105	62	59	56	61	52	64	60	94	52	49	69	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.2	31.8	31.1	29.6	27.1	26.0	26.8	29.2	31.9	31.1	31.0	31.4	(19)	
		MCWB	21.1	20.9	20.8	19.7	18.5	17.2	15.8	17.1	17.8	19.6	19.8	20.4	(20)	
	2%	DB	29.8	30.0	29.2	27.9	25.0	24.2	25.1	27.8	29.2	29.0	28.9	29.9	(21)	
		MCWB	21.5	21.3	21.0	19.7	17.8	16.6	16.2	16.6	17.8	19.4	19.6	20.2	(22)	
	5%	DB	28.1	28.8	27.8	26.1	23.0	22.7	23.2	25.9	26.8	26.2	26.9	28.1	(23)	
		MCWB	21.1	21.3	20.7	19.2	17.2	16.3	15.7	16.0	17.2	18.9	19.3	20.4	(24)	
	10%	DB	26.8	27.0	25.9	24.2	20.9	20.7	21.2	23.2	23.0	23.8	24.8	26.3	(25)	
		MCWB	20.9	21.1	20.1	18.8	16.4	15.6	15.1	15.7	17.2	18.3	18.8	20.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.7	24.1	23.3	22.2	20.3	19.0	18.7	19.6	20.5	21.8	22.2	23.1	(27)	
		MCDB	27.5	27.2	27.3	26.0	24.5	22.8	23.3	24.5	26.8	27.4	27.0	27.4	(28)	
	2%	WB	22.7	23.0	22.4	21.2	19.0	17.9	17.5	18.0	19.2	20.7	21.4	22.1	(29)	
		MCDB	26.7	26.5	26.3	24.7	22.2	21.5	22.1	24.3	25.1	25.8	25.6	26.7	(30)	
	5%	WB	22.1	22.2	21.7	20.4	18.1	17.1	16.5	17.1	18.3	19.8	20.5	21.4	(31)	
		MCDB	26.0	25.9	25.4	23.7	21.2	20.8	20.9	23.0	23.4	24.5	24.3	25.5	(32)	
	10%	WB	21.4	21.7	21.0	19.7	17.3	16.3	15.7	16.2	17.4	18.8	19.7	20.8	(33)	
		MCDB	25.0	25.2	24.2	22.7	20.1	19.5	19.6	21.3	21.7	22.4	23.1	24.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.1	9.2	9.0	9.4	9.3	10.1	10.8	11.9	10.1	9.4	9.5	9.6	(35)	
		MCDBR	11.4	11.5	11.5	11.7	12.1	13.3	14.8	16.8	16.1	13.9	12.8	12.4	(36)	
	5% WB	MCWBR	4.6	4.5	4.7	4.9	5.7	6.5	6.9	7.0	5.9	5.9	5.4	4.9	(37)	
		MCWBR	10.2	10.2	10.3	9.8	10.1	10.8	12.0	13.9	12.9	12.1	10.9	10.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.376	0.368	0.360	0.344	0.333	0.335	0.341	0.397	0.481	0.417	0.384	0.379	(40)	
		taud	2.488	2.507	2.523	2.539	2.537	2.521	2.487	2.291	2.037	2.297	2.403	2.450	(41)	
	Ebn at Noon		967	960	939	909	876	851	857	839	809	902	953	965	(42)	
		Edh at Noon	117	112	105	95	86	84	90	119	168	137	126	122	(43)	
(44) All-Sky Solar Radiation	RadAvg		5.36	5.27	4.64	3.94	3.05	2.74	2.98	3.67	3.94	4.33	5.10	5.37	(44)	
	RadStd		0.55	0.42	0.55	0.40	0.35	0.28	0.36	0.31	0.57	0.58	0.51	0.51	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	-0.51	-0.59	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	-0.51	-0.59	N/A	N/A	N/A	N/A

Nomenclature: See separate page